

Experiment 9

Logistic Regression.

Program:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Social_Network_Ads.csv')
df
df.head()

features=df.iloc[:,[2,3]].values
label=df.iloc[:,4].values

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

for i in range(1,401):
    x_train,x_test,y_train,y_test=train_test_split(
        features,label,test_size=0.2,random_state=i)
    model=LogisticRegression()
    model.fit(x_train,y_train)
    train_score=model.score(x_train,y_train)
    test_score=model.score(x_test,y_test)
    if test_score>train_score:
        print("Test {} Train{} Random State {}".format(
            test_score,train_score,i))

x_train,x_test,y_train,y_test=train_test_split(
    features,label,test_size=0.2,random_state=37)

finalModel=LogisticRegression()
finalModel.fit(x_train,y_train)

print(finalModel.score(x_train,y_train))
print(finalModel.score(x_test,y_test))

from sklearn.metrics import classification_report
print(classification_report(label,finalModel.predict(features)))
```

Output

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

400 rows × 5 columns

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4	15804002	Male	19	76000	0

Output

```
Test 0.6875 Train0.63125 Random State 3
Test 0.7375 Train0.61875 Random State 4
Test 0.6625 Train0.6375 Random State 5
Test 0.65 Train0.640625 Random State 6
Test 0.675 Train0.634375 Random State 7
Test 0.675 Train0.634375 Random State 8
Test 0.65 Train0.640625 Random State 10
Test 0.6625 Train0.6375 Random State 11
Test 0.7125 Train0.625 Random State 13
Test 0.675 Train0.634375 Random State 16
Test 0.7 Train0.628125 Random State 17
Test 0.7 Train0.628125 Random State 21
Test 0.65 Train0.640625 Random State 24
Test 0.6625 Train0.6375 Random State 25
Test 0.75 Train0.615625 Random State 26
Test 0.675 Train0.634375 Random State 27
Test 0.7 Train0.628125 Random State 28
Test 0.6875 Train0.63125 Random State 29
Test 0.6875 Train0.63125 Random State 31
Test 0.6625 Train0.6375 Random State 37
```

LogisticRegression()

0.865625

0.8125

	precision	recall	f1-score	support
0	0.85	0.93	0.89	257
1	0.84	0.71	0.77	143
accuracy			0.85	400
macro avg	0.85	0.82	0.83	400
weighted avg	0.85	0.85	0.85	400